

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 10

In this class we apply ideals to rings of polynomials.

1.1. Principal ideals in rings of polynomials.

Example 1.1. Let R be a commutative ring with unity and let $a \in R$ be any element. Then we define:

$$(1.1) \quad \langle a \rangle = \{ax \mid x \in R\}.$$

Notice that $ax + ay = a(x + y) \in \langle a \rangle$, $-ax = a(-x) \in \langle a \rangle$ and $0 = a0 \in \langle a \rangle$, so $\langle a \rangle$ is an additive subgroup of $(R, +)$. Moreover, if $y \in R$ then $yax = a(xy) \in \langle a \rangle$ so for any $ax \in \langle a \rangle$ and any $y \in R$ we have $yax \in \langle a \rangle$. That is, $\langle a \rangle$ is an ideal. This ideal is called the principal ideal generated by a .

Any ring R always have two ideals, $\{0\}$ is an ideal, and R is an ideal. We call these ideals the trivial and the improper ideal respectively.

Definition 1.1. An ideal $I \subset R$ is non-trivial and proper if $I \neq \{0\}$ and $I \neq R$.

Lemma 1.1. If R is a ring with unity and $I \subset R$ is an ideal that contains a unit $u \in I$ then $I = R$.

Proof. Let $a \in R$ be arbitrary. Then $u \cdot (a/u) \in I$ since $u \in I$ and I is an ideal. But $u \cdot (a/u) = a$ so then $a \in I$. Since a was arbitrary we conclude that $I = R$. ■

Theorem 1.1. A field F contains no proper non-trivial ideals.

Proof. Note that every $a \in F \setminus 0$ is a unit, so any ideal is either trivial or contains a unit and is therefore, by the previous lemma, all of F . ■

Definition 1.2. Let R be a ring. An ideal $M \subset R$ is a maximal ideal if M is a proper ideal that is not properly contained in a proper ideal.

Example 1.2. If p is prime then $p\mathbb{Z}$ is a proper ideal of $\mathbb{Z}$. Indeed, any ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z}$. Now, if $p\mathbb{Z} \subset n\mathbb{Z}$ then there must be some $m \in \mathbb{Z}$ such that $p = nm$ which implies that either $n = p$ and $m = 1$ or $n = 1$ and $m = p$. In the first case we would have $n\mathbb{Z} = p\mathbb{Z}$, so $p\mathbb{Z} \subset n\mathbb{Z}$ is not a proper inclusion, and in the second case we have $n\mathbb{Z} = \mathbb{Z}$ so $n\mathbb{Z}$ is not a proper ideal.

Theorem 1.2. *Let R be a commutative ring with unity. An ideal $M \subset R$ is a maximal ideal if and only if R/M is a field.*

Proof. Note that the ring homomorphism $\gamma : R \rightarrow R/M$ is surjective, so if $M \subset N$ with N an ideal then $\gamma(N) \subset R/M$ is an ideal in R/M . Now, if R/M is a field then either $\gamma(N)$ is trivial (which correspond to $N = M$) or $\gamma(N) = R/M$ (which corresponds to $N = R$).

It remains to show that if M is a maximal ideal then R/M is a field. Note that R/M is commutative (since R is) and has a unity $1+M$: $(1+M)(x+M) = (1 \cdot x) + M = x + M$. Let $x + M \in R/M$ be non-zero. Consider the ideal $\langle x \rangle + M$ (from homework this is an ideal). Since $x \notin M$ (if $x \in M$ then $x + M = 0 + M$ would be zero) it holds that M is properly contained in $\langle x \rangle + M$, so since M is maximal we conclude that $\langle x \rangle + M = R$. In particular, there must be some $a \in R$ such that $ax + m = 1$ but this implies that $1 + M = ax + M = (a + M)(x + M)$, so $x + M$ has a multiplicative inverse. Since any non-zero element in R/M has a multiplicative inverse we conclude that R/M is a field. ■

Corollary 1.2.1. *A commutative ring with unity R is a field if and only if R has no proper non-trivial ideals.*

Proof. One direction we have already shown. If R is a commutative ring with unity with no proper non-trivial ideals then $\{0\}$ is a maximal ideal so $R \cong R/\{0\}$ is a field. ■

Definition 1.3. *An ideal $P \subset R$ is a prime ideal if $ab \in P$ implies that either $a \in P$ or $b \in P$.*

Theorem 1.3. *Let R be a commutative ring with unity and $P \subset R$ an ideal. Then R/P is an integral domain if and only if P is a prime ideal.*

Proof. We already know that R/P is a commutative ring with unity, where the unity is given by $1+P$. If R/P is an integral domain and $ab \in P$ then $(a+P)(b+P) = 0+P$ which implies that either $a + P = P$ or $b + P = P$ which in turn implies $a \in P$ or $b \in P$.

Conversely, if P is a prime ideal and if $(a + P)(b + P) = 0 + P$ then $ab + P = P$ or $ab \in P$. Using that P is a prime ideal we conclude that $a \in P$ or $b \in P$ which implies $a + P = 0$ or $b + P = 0$. ■

Theorem 1.4. *If $I \subset F[x]$ is an ideal then $I = \langle f(x) \rangle$ for some $f(x) \in F[x]$.*

Proof. If $I = \{0\}$ then $I = \langle 0 \rangle$ and we are done, so assume that $I \neq \{0\}$. Let $f(x) \in I$ be a non-zero polynomial with minimal degree in I . Let $g(x) \in I$, then we can write $g(x) = q(x)f(x) + r(x)$ with $\deg(r(x)) < \deg(f(x))$. Since $g(x), f(x) \in I$ and since I is an ideal we have $r(x) = g(x) - q(x)f(x) \in I$. We conclude that $r(x) = 0$ since otherwise $\deg(f(x)) > \deg(r(x))$ would contradict our choice of $f(x)$ in I . It follows that $g(x) = q(x)f(x) \in \langle f(x) \rangle$, so since $g(x) \in I$ was arbitrary we conclude that $I = \langle f(x) \rangle$. ■

Theorem 1.5. *An ideal $\langle f(x) \rangle \subset F[x]$ is a maximal ideal if and only if $f(x)$ is irreducible.*

Proof. Assume that $\langle f(x) \rangle$ is maximal. In this case $\langle f(x) \rangle$ is also a prime ideal, so if $f(x) = g(x)h(x)$ then $g(x) \in \langle f(x) \rangle$ or $h(x) \in \langle f(x) \rangle$. Since any (non-zero) polynomial in $\langle f(x) \rangle$ has degree at least $\deg(f(x))$ we conclude that either $\deg(h(x))$ or $\deg(g(x))$ has degree $\deg(f(x))$. But this implies that $f(x)$ is irreducible.

Conversely, assume that $f(x)$ is irreducible and let $\langle f(x) \rangle \subset N \subset R[x]$ with N some ideal. We find $g(x)$ such that $N = \langle g(x) \rangle$, so since $f(x) \in \langle f(x) \rangle \subset \langle g(x) \rangle$ so $f(x) = q(x)g(x)$. If $g(x)$ is constant then $N = R[x]$, and if $q(x)$ is constant then $N = \langle af(x) \rangle = \langle f(x) \rangle$, so we see that $\langle f(x) \rangle$ is not properly contained in a larger proper ideal, so $\langle f(x) \rangle$ is maximal. ■

Theorem 1.6. *If $f(x) \in F[x]$ is irreducible and $f(x)$ divides $g(x)h(x)$ then $f(x)$ divides either $g(x)$ or $h(x)$.*

Proof. If $g(x)h(x) = q(x)f(x)$ then $g(x)h(x) \in \langle f(x) \rangle$. Since $\langle f(x) \rangle$ is maximal and therefore prime it follows that either $g(x) \in \langle f(x) \rangle$ or $h(x) \in \langle f(x) \rangle$ which implies that $g(x) = p(x)f(x)$ or $h(x) = r(x)f(x)$. ■

Email address: sandfeldt@uchicago.edu